Amprius Technologies (AMPX)

Silicon-Anode Scaling Is the Real Value Driver, Not the Near-Term Price Spike

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> Amprius Technologies is positioned to capture a growing share of the high-energy battery market through its silicon-anode technology, yet the market has over-reacted to short-term pricing volatility and ignored the longer-term compounding of margin expansion. While consensus estimates assume a flat trajectory, we see a clear path to margin improvement and revenue acceleration once the technology reaches volume production. Accordingly, we assign a Hold rating with valuation concerns, a blended target of $5.64 and a -30.5% upside from the current $8.12 price.

HOLD	$5.64	-30.5%	$8.12
Hold — Valuation Concerns	BLENDED TARGET	BLENDED UPSIDE	REFERENCE PRICE
INVESTMENT RATING			

The market is fixated on the next quarterly earnings beat, while the true investment thesis unfolds over a multi-year horizon as Amprius scales silicon-anode cells and secures strategic OEM contracts.

EXECUTIVE SUMMARY

Amprius operates a vertically integrated battery platform that combines proprietary silicon-anode materials with advanced cell engineering. The company’s revenue is currently concentrated in niche high-energy applications such as premium electric vehicles, aerospace, and specialty consumer electronics. In 2025 the firm captured roughly 0.3% of the global lithium-ion market, a share that is expanding as OEMs seek higher specific energy to meet range targets. The technology advantage lies in a 30% higher energy density relative to conventional graphite anodes, which translates into lighter packs and longer range for vehicle manufacturers. The recent market overreaction stems from a combination of a temporary supply-chain bottleneck and a sharp swing in raw-material pricing that compressed Amprius’s gross margin in Q2 2026. The resulting earnings miss triggered a sell-off that pushed the stock 22% lower in two weeks, creating a valuation gap that our discounted cash flow model captures. Our baseline scenario assumes a 12% compound annual revenue growth rate as the company ramps volume production in its new Arizona fab, while normalizing gross margins to 38% by 2029. Capital allocation remains disciplined. Amprius has earmarked $120 million for fab expansion, while maintaining a lean balance sheet with $85 million of cash and $45 million of revolving credit. The firm continues to invest roughly 18% of revenue in R&D, focusing on next-generation silicon-nanowire architectures that promise further energy density gains. No material M&A activity is anticipated in the near term, allowing management to preserve cash for organic growth.

OPERATIONAL SNAPSHOT

$8.12	$48.3 million	$5.3 million	11.0%	1.30	$8.12
REF. PRICE	LTM REVENUE	LTM FREE CASH FLOW	FCF MARGIN	EQUITY BETA	REF. CLOSE

LTM figures are Cascadia estimates grounded in the latest filings; ref. price as of 2026-10-14.

THE MOAT IS DEEP AND SUSTAINABLE

Amprius's silicon-anode platform creates a switching cost that is difficult for competitors to replicate. The proprietary nanowire synthesis process requires multi-step high-temperature annealing that is protected by a portfolio of 27 patents covering material composition, electrode architecture, and cell integration. Customers that have qualified the technology for automotive use must undergo extensive validation cycles, which locks in revenue for the duration of the vehicle program. Regulatory barriers further reinforce the moat. The U.S. Department of Energy's Advanced Battery Consortium has designated silicon-anode cells as a strategic technology, granting Amprius preferential access to federal grants and fast-track approvals for safety testing. This regulatory tailwind reduces the time to market for new designs and raises the hurdle for new entrants that lack similar government relationships. Pricing power emerges from the high-energy advantage. OEMs are willing to pay a premium of up to 15% for a 30% increase in specific energy because it directly translates into vehicle range and weight savings. As the technology matures, the premium is expected to compress modestly, but the underlying value proposition remains intact, supporting a resilient gross margin profile. Finally, the company's long-term contracts with tier-one automotive suppliers embed a revenue runway that extends beyond the next product cycle. These agreements include volume-based rebates that incentivize Amprius to deliver higher yields, aligning shareholder interests with customer success.

VALUATION

Three scenarios make the valuation range explicit

Our discounted cash flow framework evaluates three distinct pathways for revenue growth, margin evolution, and terminal assumptions. The baseline case assumes a 12% revenue CAGR through 2035, a gradual lift in free cash flow margin to 14% as scale drives cost efficiencies, and a terminal growth rate of 3.0% reflecting the long-term expansion of the electric-vehicle market. The bear case reduces revenue CAGR to 5% and caps the free cash flow margin at 5% due to slower fab ramp-up and heightened competition. The bull case envisions a 20% CAGR driven by early adoption in premium EVs and a free cash flow margin of 18% as the company achieves full-scale production.

SCENARIO	WEIGHT	REVENUE CAGR	TARGET FCF MARGIN	TARGET	UPSIDE
Bear	25%	5%	5%	$2.15	-73.5%

Base	50%	12%	14%	$5.12	-36.9%
Bull	25%	20%	18%	$10.19	+25.5%

Bear $2.15

In this outcome the Arizona fab experiences prolonged yield issues, raw-material costs remain elevated, and a new entrant captures a share of the premium EV market. The combination of lower volumes and compressed margins drives free cash flow to a fraction of current levels.

Base $5.12

The most likely path sees the fab achieving 80% capacity by 2029, gross margins normalizing to 38%, and a steady pipeline of OEM contracts delivering 12% annual revenue growth. Free cash flow improves as capex declines relative to revenue.

Bull $10.19

For the bull case the company secures a flagship partnership with a leading EV manufacturer, accelerates volume production, and leverages its patent portfolio to command a premium pricing regime. Margin expansion and rapid top-line growth combine to lift valuation well above current levels.

VALUATION FRAMEWORK

A live discount rate, not a static hurdle

We construct the discount rate using a CAPM approach that incorporates the 1.30 equity beta, a risk-free rate of 4.2%, and an equity risk premium of 6.5%. Adding a 1.5% size premium yields a cost of equity of 12.2%. After accounting for the company's modest debt load, the weighted average cost of capital settles at 11.38% as of 2026-10-14. This live rate reflects the market's pricing of Amprius's technology risk, execution risk, and the broader macro environment. The 11.38% WACC implies that investors require a relatively high return to compensate for the nascent stage of silicon-anode adoption. Consequently, any upside in free cash flow generation translates into a proportionally larger impact on intrinsic value, reinforcing the importance of margin normalization in our model.

FCF margin normalization is the engine

Our analysis assumes that free cash flow margins will improve as the company transitions from a pilot-scale operation to a mature, high-volume fab. In the base scenario, gross margins rise from 32% in 2025 to 38% by 2029, while operating expenses scale sub-linearly, delivering a free cash flow margin of 14% at the end of the forecast horizon. The bear scenario holds margins near current levels, reflecting persistent yield challenges, whereas the bull scenario pushes margins to 18% as the firm captures pricing power and benefits from economies of scale. Margin dynamics are central to valuation because they dictate the cash available

KEY VALUATION PARAMETERS

Revenue base year	**2026**
Forecast horizon	**10 years**
Live WACC	**11.38% (2026-10-14)**
Terminal growth	**3.0%**
Equity beta	**1.30**
Diluted shares	**200 million**

HOLD

over $5.64 blended target

for reinvestment, debt reduction, and shareholder returns. The DCF model therefore places heavy weight on the trajectory of free cash flow margins across the three scenarios.

Mechanics are explicit

Revenue becomes EBIT, cash taxes come out, then we add back D&A and stock compensation, subtract CapEx and the change in net working capital to reach unlevered free cash flow, which we discount and divide by diluted shares.

Methodology. Detailed EBIT-bridge DCF on three scenarios, probability-weighted 25/50/25. Beta decays toward a mature peer over the forecast. Terminal value via Gordon growth. Targets are model-computed, not estimated.

BUSINESS & SEGMENT DRIVERS

Revenue prospects anchor on Silicon-Anode Scaling and OEM Partnership Expansion

HOLD

The silicon-anode segment currently contributes roughly 68% of total revenue, driven by high-energy cells sold to premium EV makers. As the Arizona fab reaches 80% utilization, we anticipate a 15% lift in segment revenue each year, supported by incremental capacity and improved yields. The remaining 32% of revenue stems from aerospace and specialty consumer electronics, where the technology’s weight advantage commands a premium price. OEM partnership expansion is the second growth lever. Amprius has signed memoranda of understanding with three Tier-1 automotive suppliers, each representing a potential $30 million annual revenue stream once volume production commences. The contracts include performance milestones that trigger additional supply commitments, creating a step-function in top-line growth as each milestone is achieved. What to monitor

- **Fab yield trajectory** The speed at which the Arizona facility achieves target yields directly influences margin expansion and cash generation.
- **OEM contract conversion** The transition from pilot agreements to full-scale supply contracts will validate the revenue forecasts.
- **Raw-material pricing** Silicon feedstock costs remain a material input; sustained price pressure could compress gross margins.
- **Regulatory approvals** Safety certifications for high-energy cells in the U.S. and EU affect market access and timing.

12% YoY	11.0%	38%	18%	7%	$85 million
REVENUE GROWTH	FCF YIELD	GROSS MARGIN	R&D INTENSITY	CAPEX % REV	CASH BALANCE

METRIC	2025
Revenue	$48.3 million
Gross Profit	$18.4 million
EBIT	$5.2 million
Net Income	$2.1 million
Free Cash Flow	$5.3 million
R&D	$8.7 million
ROE	4.9%

RISK & MOAT MATRIX

The bear case

SHARE EROSION

Prolonged yield issues could force Amprius to miss volume targets, prompting customers to shift to alternative chemistries and diluting the share price.

M&A DIGESTION

A future acquisition aimed at scaling capacity may not integrate smoothly, leading to higher integration costs and distraction from core execution.

REGULATORY AND PRICING SCRUTINY

Heightened safety scrutiny in the EU could delay certification, while price caps on high-energy cells would compress the premium pricing advantage.

TECHNOLOGY DISRUPTION

Breakthroughs in solid-state batteries could render silicon-anode advantages less compelling, eroding the competitive edge.

REINVESTMENT TREADMILL

Continuous capital needs for fab upgrades may consume free cash flow, limiting the ability to return capital to shareholders.

WACC AND DURATION SENSITIVITY

Rising interest rates would increase the discount rate, disproportionately lowering the present value of long-term cash flows.

The strengths

OLIGOPOLY POSITION

Amprius sits among a handful of firms that have

FCF COMPOUNDING

As scale drives margin expansion, free cash flow

GEOGRAPHIC DURABILITY

PIPELINE OPTIONALITY

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successfully commercialized silicon-anode cells, creating a de-facto oligopoly that limits competitive pressure.

is expected to compound at a faster rate than revenue, providing a powerful engine for shareholder value.

Strategic location of the Arizona fab near key automotive supply chains reduces logistics costs and shields the operation from geopolitical supply shocks.

Beyond automotive, the company's roadmap includes high-energy cells for drones and grid-scale storage, offering multiple avenues for growth.

Compelling risk-reward, skewed to the upside

Our analysis indicates that the market has priced in a near-term earnings miss without fully appreciating the long-run cash-flow upside embedded in silicon-anode scaling. While execution risk remains, the combination of a deep patent moat, strategic OEM partnerships, and a disciplined capital structure supports a favorable risk-reward profile. Accordingly, we maintain a Hold rating with valuation concerns, a blended target of $5.64 and a -30.5% upside relative to the current $8.12 price.

Justin Wen, Senior Equity Research Analyst, Cascadia Analytics

FRANCHISE DEEP DIVE

Silicon-Anode Cells: repriced on normalization

The flagship silicon-anode cell line, marketed under the "SiliconX" brand, delivers 30% higher specific energy than conventional graphite cells. Early shipments to premium EV makers were priced at a 20% premium, reflecting the novelty of the technology and limited production capacity. As the Arizona fab ramps to 80% utilization, unit costs are expected to fall by 12% annually, allowing the premium to normalize to roughly 8% by 2029. This price compression is offset by higher volumes and improved yields, which together lift free cash flow margins. The growth narrative for SiliconX initially hinged on rapid adoption across the mass-market EV segment. However, the technology's current manufacturing complexity has constrained that ambition, leading to a short-term revenue shortfall. The market's reaction was therefore disproportionate, as investors focused on the immediate premium erosion rather than the underlying cost-curve improvements.

Why the growth numbers disappointed

- Yield ramp-up at the Arizona fab lagged the internal schedule by six months, reducing shipped volume in Q2 2026.
- Raw-material price spikes for high-purity silicon increased cost of goods sold, compressing gross margin.
- OEM customers deferred full-scale orders pending validation of long-term cycle life under real-world conditions.
- Regulatory certification delays in the EU added an additional quarter before market entry.

Our read: Normalization, not a thesis break. The premium pricing is receding as scale improves, yet the underlying technology advantage remains intact. As the company moves from pilot to volume production, cash-flow generation will accelerate, supporting a valuation that is materially higher than current market levels.